SUPPORTING ADULT GUIDANCE

Right Triangles and Hidden Squares

Pythagoras' Theorem. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner builds Pythagoras' Theorem from first principles by comparing the areas of squares drawn on the three sides of a right-angled triangle. The emphasis is on **seeing an invariant relationship** — the two smaller square areas always sum to the largest — and expressing it, first in words and then as $a^2 + b^2 = c^2$.

Driving Question: *What can three squares reveal about one right-angled triangle?*

The mathematics, in plain terms

For any right-angled triangle, if you build a square on each side, the areas obey $a^2 + b^2 = c^2$, where c is the hypotenuse (the longest side, opposite the right angle). "The square on a side" means the area of a square whose side is that length — so a^2 literally is that area. The whole lesson is about noticing that the two smaller square areas together equal the square area on the hypotenuse, whatever triangle you draw.

What makes the relationship trustworthy

The reason to drag many different triangles is that a pattern seen once might be a coincidence; a pattern that survives every change you try is a genuine relationship. Encouraging a learner to predict before revealing, and to test a few deliberately different triangles, is what turns "it worked that time" into "it always works". That habit — looking for what stays the same — is the heart of the lesson.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The three squares are connected: two areas combine to make the third. The hypotenuse connects to the right angle opposite it.	~15 min
Movement	Dragging the triangle and watching the squares change — the relationship moves but survives.	~12 min
Reflection	Asking: does the same relationship hold every time? What exactly stays the same?	~7 min
Creativity	Making a card or picture that shows how three squares become one formula.	~4 min
Rest	The pause to predict before revealing a calculation — space before proof.	~3 min

Nutrition	Intellectual (Mode B): this lesson feeds pattern-awareness and the pleasure of seeing hidden structure.	~4 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Comparing side lengths instead of square areas

The learner adds the side lengths and expects the third side. Repair: “We are comparing the *areas* of the squares, not the sides. What is the area of this square?”

Expecting the relationship to change with the triangle

Repair: “Try a very different right-angled triangle. What has changed — and what has stayed exactly the same?”

Not knowing which square sits on the hypotenuse

Repair: “Which side is opposite the right angle? The square on that side is the largest — which of the three is it?”

Confusing the capital vertex with the lower-case side

Repair: “Capital A is a corner; lower-case a is the length of the side opposite it. Which is which here?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: square numbers, square roots, and right angles. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

This lesson is about noticing, so the most useful support is questions that send the learner back to look:

- “Before you reveal anything — do you think the two smaller squares together will be bigger than, smaller than, or the same as the big one?”
- “What is the area of this square? And this one? Add them — now look at the big square.”
- “Try another triangle. Did the relationship survive?”
- “How could you say what you have found in one sentence, without any numbers?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise the formula today?”
No. Today is about *seeing* the relationship and saying it in words. The symbols $a^2 + b^2 = c^2$ grow naturally out of that; memorising comes later, with understanding underneath it.

“Why squares, and not just the numbers?”
The squares make the relationship visible — you can literally see two areas fitting into a third. It is much harder to believe from numbers alone.

“Their triangle looks different from the example — is that a problem?”
Not at all — that is the point. The relationship is meant to survive every right-angled triangle, so different triangles are welcome evidence.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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